

MINI PROJECT REPORT

LOAN DEFAULT & CREDIT RISK ANALYSIS

1. PROJECT TITLE

Loan Default & Credit Risk Analysis

2. PROJECT OBJECTIVE

The objective of this project is to analyze customer loan information and identify patterns associated with loan default.

The project analyzes customer demographics, annual income, credit score, loan amount, interest rate, debt-to-income ratio, previous defaults, employment status, loan type, and other financial factors.

Python and Power BI are used to clean, analyze, visualize, and present the data in an understandable form.

The analysis helps understand credit-risk patterns and provides useful information for financial risk monitoring and decision-making.

3. DATASET DESCRIPTION

The dataset contains customer and loan-related information.

The dataset contains **500 records and 24 columns**.

Important columns include:

- Customer_ID
- Gender
- Age
- Education
- Employment_Status

- Marital_Status
 - Dependents
 - Annual_Income
 - Credit_Score
 - Existing_Loans
 - Loan_ID
 - Loan_Type
 - Loan_Amount
 - Loan_Term_Months
 - Interest_Rate
 - Monthly_Installment
 - Debt_to_Income_Ratio
 - Employment_Years
 - Previous_Defaults
 - Credit_History
 - Property_Ownership
 - Region
 - Loan_Status
 - Default_Status
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4. TOOLS AND TECHNOLOGIES USED

The following tools were used in this project:

- Python
- Pandas
- NumPy

- Matplotlib
 - Seaborn
 - Google Colab
 - Power BI Desktop
 - Power Query
 - DAX
 - GitHub
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5. DATA LOADING

The loan dataset was loaded into Python using the Pandas library.

The first five and last five records were displayed to understand the structure of the dataset.

The dataset shape and column names were also checked.

Dataset Size

- Rows: **500**
 - Columns: **24**
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6. DATA EXPLORATION AND CLEANING

The dataset was explored and cleaned before performing analysis.

The following steps were performed:

6.1 Data Types

The data types of all columns were checked using Pandas.

6.2 Statistical Information

Statistical information such as count, mean, minimum, maximum, and standard deviation was analyzed for numerical columns.

6.3 Missing Values

The dataset was checked for missing values.

Result: No significant missing values were found in the cleaned dataset.

6.4 Duplicate Records

Duplicate records were checked.

Result: No duplicate records were found.

6.5 Unique Values

Unique values were checked for categorical columns such as:

- Gender
- Education
- Employment Status
- Loan Type
- Region
- Default Status

6.6 Outlier Checking

Numerical columns such as loan amount, annual income, credit score, and debt-to-income ratio were checked for unusual values.

6.7 Final Dataset

After cleaning and verification, the dataset was used for further analysis and visualization.

7. DATA ANALYSIS

The following analysis was performed:

Customer and Loan Analysis

- Total customers
- Total loans
- Defaulted loans

- Non-defaulted loans
- Overall default rate

Financial Analysis

- Average annual income
- Average credit score
- Average loan amount
- Average interest rate
- Average debt-to-income ratio

Category Analysis

- Number of loans by loan type
- Number of loans by region
- Default rate by loan type
- Default rate by employment status
- Default rate by education
- Default rate by property ownership
- Default rate by credit-score group
- Default rate by income group

Risk Analysis

- Customers below average credit score
- Customers above average loan amount
- Customers with high debt-to-income ratio
- Top 10 customers based on loan amount
- Loan type with highest default rate
- Region with highest default rate
- Credit score vs default

- Income vs default
 - Loan amount vs default
 - Previous defaults vs current default
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8. IMPORTANT ANALYSIS RESULTS

Some important results from the analysis are:

Metric	Result
Total Customers	500
Total Loans	500
Defaulted Loans	276
Non-Defaulted Loans	224
Overall Default Rate	55.2%
Average Credit Score	685.49
Average Interest Rate	12.64%
Average DTI Ratio	37.77%

These values help provide an overall understanding of the loan portfolio and credit-risk situation.

9. PYTHON VISUALIZATIONS

The following visualizations were created using Matplotlib and Seaborn:

1. Loan Default Distribution – Pie Chart
2. Loan Count by Loan Type – Bar Chart
3. Default Rate by Loan Type – Bar Chart
4. Default Rate by Employment Status – Bar Chart

5. Default Rate by Education – Bar Chart
6. Credit Score Distribution – Histogram
7. Loan Amount Distribution – Histogram
8. Annual Income Distribution – Histogram
9. Credit Score vs Loan Amount – Scatter Plot
10. Income vs Loan Amount – Scatter Plot
11. Loan Amount by Default Status – Box Plot
12. Credit Score by Default Status – Box Plot
13. Debt-to-Income Ratio by Default Status – Box Plot
14. Default Rate by Income Group – Bar Chart
15. Correlation Heatmap

These visualizations make it easier to identify patterns and relationships in the loan data.

10. POWER BI DASHBOARD

A Power BI dashboard titled:

"Loan Default & Credit Risk Analytics Dashboard"

was created to provide an interactive view of the loan data.

KPI Cards

The dashboard contains the following KPI cards:

- Total Customers
- Total Loans
- Defaulted Loans
- Default Rate %
- Average Loan Amount

- Average Credit Score
- Average Annual Income

Power BI Visuals

The dashboard contains:

1. Loan Default Distribution – Donut Chart
2. Default Rate by Loan Type – Bar Chart
3. Default Rate by Employment Status – Bar Chart
4. Default Rate by Credit Score Group – Column Chart
5. Default Rate by Income Group – Bar Chart
6. Loan Amount by Loan Type – Column Chart
7. Average Loan Amount by Region – Bar Chart
8. Credit Score Distribution – Histogram
9. Income vs Loan Amount – Scatter Chart
10. Loan Details – Table/Matrix
11. Default Distribution – Appropriate Chart

Power BI Slicers

Interactive slicers were added for:

- Gender
- Education
- Employment Status
- Loan Type
- Region
- Default Status
- Credit Score Group

These slicers allow users to filter and explore the dashboard interactively.

11. DAX MEASURES

The following DAX measures were created:

Total Customers

Total Customers =

DISTINCTCOUNT('loan_default_credit_risk_analysis_cleaned'[Customer_ID])

Total Loans

Total Loans =

DISTINCTCOUNT('loan_default_credit_risk_analysis_cleaned'[Loan_ID])

Defaulted Loans

Defaulted Loans =

CALCULATE(
 [Total Loans],
 'loan_default_credit_risk_analysis_cleaned'[Default_Status] = "Default"
)

Non-Defaulted Loans

Non-Defaulted Loans =

CALCULATE(
 [Total Loans],
 'loan_default_credit_risk_analysis_cleaned'[Default_Status] = "No Default"
)

Default Rate %

Default Rate % =

DIVIDE([Defaulted Loans], [Total Loans], 0)

Average Loan Amount

Average Loan Amount =

AVERAGE('loan_default_credit_risk_analysis_cleaned'[Loan_Amount])

Average Credit Score

Average Credit Score =

AVERAGE('loan_default_credit_risk_analysis_cleaned'[Credit_Score])

Average Annual Income

Average Annual Income =

AVERAGE('loan_default_credit_risk_analysis_cleaned'[Annual_Income])

12. BUSINESS INSIGHTS

1. What is the overall loan default rate?

The overall loan default rate is **55.2%**. This means that 55.2% of the loans in the dataset are classified as defaulted.

2. Which loan type has the highest default rate?

Auto Loans have the highest default rate, at approximately **60%**.

3. Which region has the highest default rate?

The **Central region** has the highest default rate, at approximately **61.21%**.

4. Does credit score appear associated with loan default?

Yes. The analysis shows that customers with **lower credit scores generally have higher default risk** compared with customers having higher credit scores.

5. Does income appear associated with default?

Default rates vary between income groups. Income therefore shows differences in default patterns across customer segments.

6. Are higher debt-to-income ratios associated with default?

Customers with **higher debt-to-income ratios generally show greater default risk**. A high DTI indicates that a larger portion of income may already be committed to debt obligations.

7. Are previous defaults associated with current default?

Yes. Customers with **previous defaults show higher current default risk** compared with customers without previous defaults.

8. Which customer segment has the highest default rate?

Customer segments with **higher-risk financial characteristics**, such as lower credit scores and higher debt-to-income ratios, show higher default patterns.

9. Which factors appear associated with loan default?

Important factors associated with default patterns include:

- Credit Score
- Previous Defaults
- Debt-to-Income Ratio
- Loan Type
- Employment Status
- Income
- Region
- Loan Amount

10. What patterns can be observed between loan amount and default status?

Loan amounts vary between defaulted and non-defaulted customers. Loan amount should therefore be considered together with income, credit score, and debt-to-income ratio when analyzing credit risk.

11. What financial factors should an organization monitor?

An organization should monitor:

- Credit Score
- Annual Income

- Loan Amount
 - Interest Rate
 - Debt-to-Income Ratio
 - Previous Defaults
 - Existing Loans
 - Credit History
 - Monthly Installment
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13. KEY BUSINESS INSIGHTS

The major findings from the project are:

1. The overall loan default rate is **55.2%**.
 2. **Auto Loans** have the highest default rate among the loan types.
 3. The **Central region** has the highest default rate.
 4. Lower credit scores are associated with higher default risk.
 5. Higher debt-to-income ratios are associated with increased default risk.
 6. Previous defaults are an important indicator of current default patterns.
 7. Income groups show differences in default rates.
 8. Loan amount should be analyzed together with income and credit score.
 9. Credit score, DTI ratio, previous defaults, and loan characteristics are important factors for credit-risk monitoring.
 10. The analysis can help identify customer segments that require closer credit-risk monitoring.
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14. CONCLUSION

The **Loan Default & Credit Risk Analysis** project successfully analyzed customer and loan data using Python and Power BI.

The project included data loading, data cleaning, exploratory analysis, statistical analysis, visualization, DAX calculations, and an interactive Power BI dashboard.

The analysis identified important patterns related to loan defaults, including credit score, debt-to-income ratio, previous defaults, loan type, income, and region.

The Power BI dashboard provides an interactive way to explore these patterns using KPIs, charts, tables, and slicers.

Overall, this project demonstrates how data analytics can be used to understand loan performance and support **credit-risk monitoring and business decision-making**.

15. PROJECT SUBMISSION FILES

The final project contains:

StudentName_Loan_Credit_Risk_Analysis

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|— Dataset

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| — Loan_Credit_Risk_Cleaned.csv

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|— Python

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| — Loan_Credit_Risk_Analysis.ipynb

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|— PowerBI

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| — Loan_Credit_Risk_Dashboard.pbix

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|— Report

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| — Loan_Credit_Risk_Project_Report.pdf

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└─ README.md

16. FINAL CHECKLIST

- ☒ Dataset loaded
- ☒ Data cleaned
- ☒ Missing values checked
- ☒ Duplicate values checked
- ☒ Python analysis completed
- ☒ Python visualizations completed
- ☒ Power BI data model completed
- ☒ DAX measures created
- ☒ KPI cards created
- ☒ Charts created
- ☒ Slicers added
- ☒ Dashboard formatted
- ☒ Business insights written
- ☒ Project report completed
- ☒ Python file submitted
- ☒ Cleaned dataset submitted
- ☒ PBIX file submitted
- ☒ PDF report submitted
- ☒ README completed